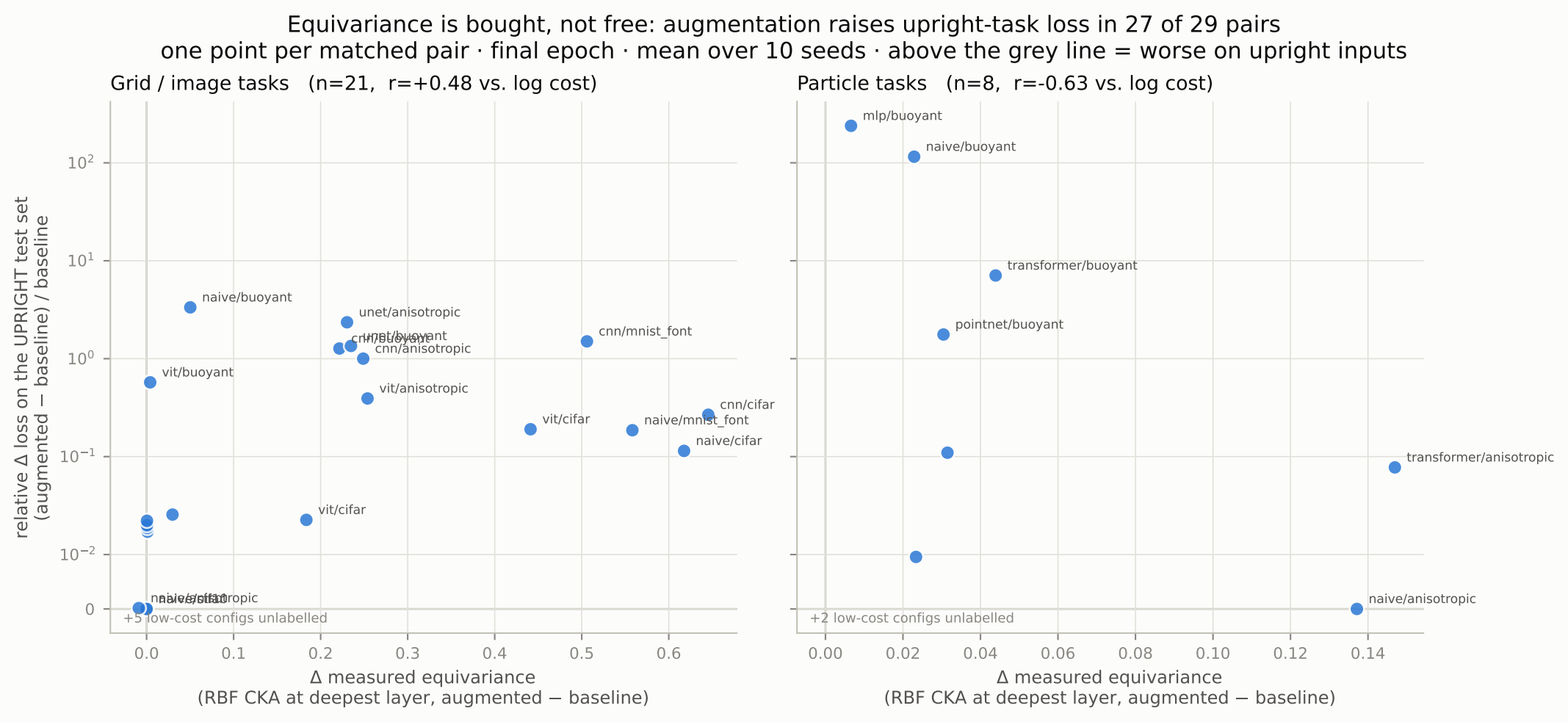

Equivariance is bought, not free: augmentation raises upright-task loss in 27 of 29 pairs
one point per matched pair · final epoch · mean over 10 seeds · above the grey line = worse on upright inputs
Grid / image tasks (n=21, r=+0.48 vs. log cost)
Particle tasks (n=8, r=-0.63 vs. log cost)
relative Δ loss on the UPRIGHT test set
(augmented − baseline) / baseline
naive/buoyant
unet/anisotropic
cnn/buoyant
unet/buoyant
cnn/mnist_font
cnn/anisotropic
vit/buoyant
vit/anisotropic
cnn/cifar
vit/cifar
naive/mnist_font
naive/cifar
vit/cifar
+5 low-cost configs unlabelled
Δ measured equivariance
(RBF CKA at deepest layer, augmented − baseline)
mlp/buoyant
naive/buoyant
transformer/buoyant
pointnet/buoyant
transformer/anisotropic
naive/anisotropic
+2 low-cost configs unlabelled
Δ measured equivariance
(RBF CKA at deepest layer, augmented − baseline)